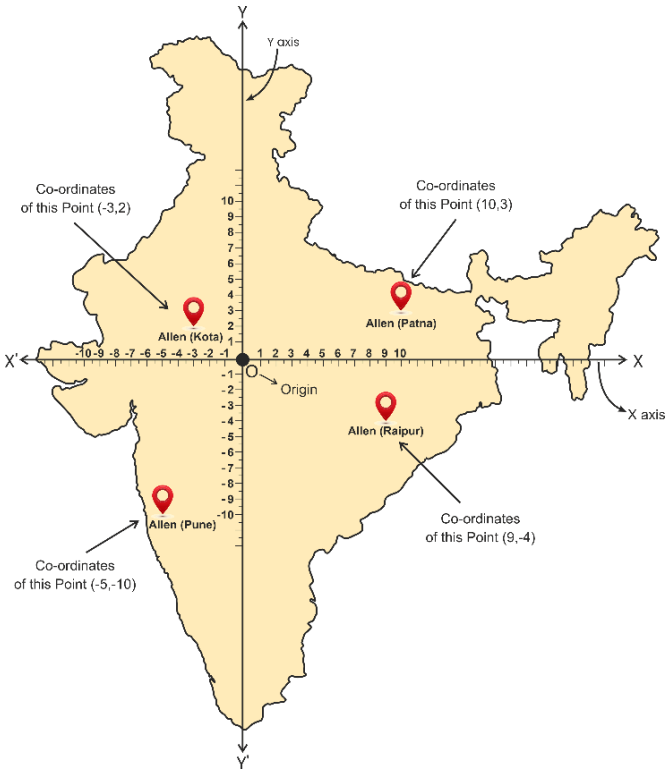
5 Co-ordinate Geometry



Cartesian coordinate system

Cartesian coordinate axes

On a graph paper two mutually perpendicular straight lines x'ox and yoy', intersecting each other at point o.

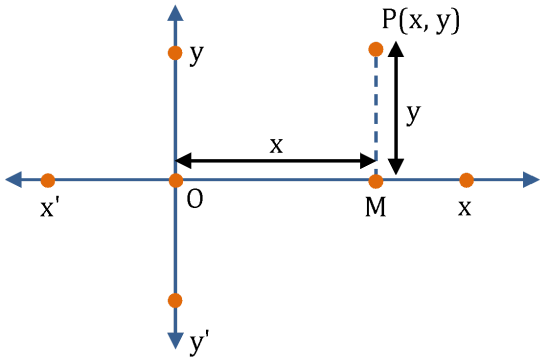
The horizontal line x'ox is called x-axis and the vertical line yoy' is called y-axis and the two lines x'ox and yoy' taken together are known as **coordinate axes** or **rectangular axes** or **axes** or **reference**. The point 'o' is called origin.

Let us assume that two perpendicular number lines x'ox and yoy' intersect each other at the point o. The horizontal number line is called the x-axis and the vertical number line is called y-axis. The point o, where these lines intersect, is called the origin (the zero point).

We choose a unit length and mark the positive integers 1, 2, 3,.....on the right of the origin on x-axis and above the origin on y-axis. Negative integers are marked to the left of origin on x-axis and below the origin on y-axis.

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Note : The cartesian coordinate system or the rectangular system was invented by French mathematician Rene Descartes when he tried to describe the path of a fly crawling along crisscross brans on the ceiling while he lay on his bed.



The above system is called a cartesian, or rectangular coordinate system.

A pair of two numbers x and y, listed in a specific order, is called an ordered pair (x, y), with x at the first place and y at the second place.

Be Alert !

- Note that **(x, y) ≠ (y, x)**

Do You Remember ?

- In our earlier classes, we studied about cartesian coordinate system. We have seen that a 2-D plane is divided by 2 number lines (intersecting at 90°) into 4 parts or quadrants.

  The position of point on a graph paper was shown by pair of values called "Coordinates".

  In this chapter, we will continue plotting the points. Observe some shapes, try to find their areas and even try to locate the reflection of points with respect to coordinate axis.

Building Concepts 1

Consider an equation y = 2x + 1 where x = 0, 1, 2.

Write the ordered pairs specifying (x, y).

Explanation

We have y = 2x + 1

When x = 0, y = 2 × 0 +1 = 1

When x = 1, y = 2 × 1 + 1 = 3

When x = 2, y = 2 × 2 + 1 = 5

The ordered pairs are (0, 1), (1, 3) and (2, 5).

Coordinate of a point (or cartesian coordinates)

On a graph paper, let x'ox and y'oy be the coordinate axes. Let P be a point on this graph paper.

Let P be at a distance of

(i) a units from the y-axis.

(ii) b units from the x-axis.

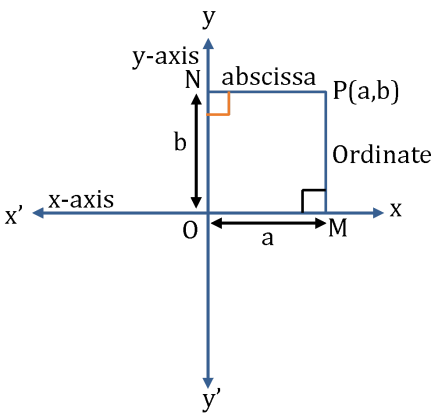
Then, we say that the coordinates of P are (a, b).

a is called the x-coordinate or abscissa of P

b is called the y-coordinate or ordinate of P

(a, b) is also called the cartesian coordinates of the point P

The sign of coordinates are different in the different quadrants.

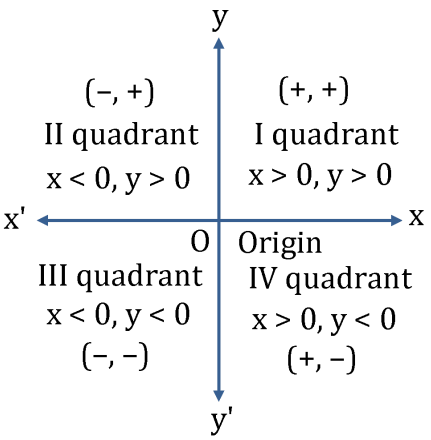


Quadrants

The coordinate axes (plural of axis) divide the plane of the graph paper into four regions, called quadrants.

Using the convention of signs, we have:

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Region	Quadrant	Nature of x and y	Sign of the coordinates
xoy	I	x > 0 and y > 0	(+, +)
yox'	II	x < 0 and y > 0	(–, +)
x'oy'	III	x < 0 and y < 0	(–, –)
y'ox	IV	x > 0 and y < 0	(+, –)

(i) Both the coordinates are positive in the Ist quadrant.
(ii) Abscissa is negative and ordinate is positive in the IInd quadrant.
(iii) Both abscissa and ordinate are negative in the IIIrd quadrant.
(iv) Abscissa is positive and ordinate is negative in the IVth quadrant.

Coordinate of a point in the cartesian plane

Consider a point, say P in the first quadrant. We can reach P by moving 4 units along OX and then moving 5 units upward parallel to OY. The point P is read as P (4, 5). Here, 4 is the x-coordinate and 5 is the y-coordinate. The x-coordinate is also called abscissa and the y- coordinate as the ordinate of the point. Thus, every point on the coordinate plane has its abscissa and ordinate. The abscissa and the ordinate together are called the coordinates of the point.

Distance between two points:

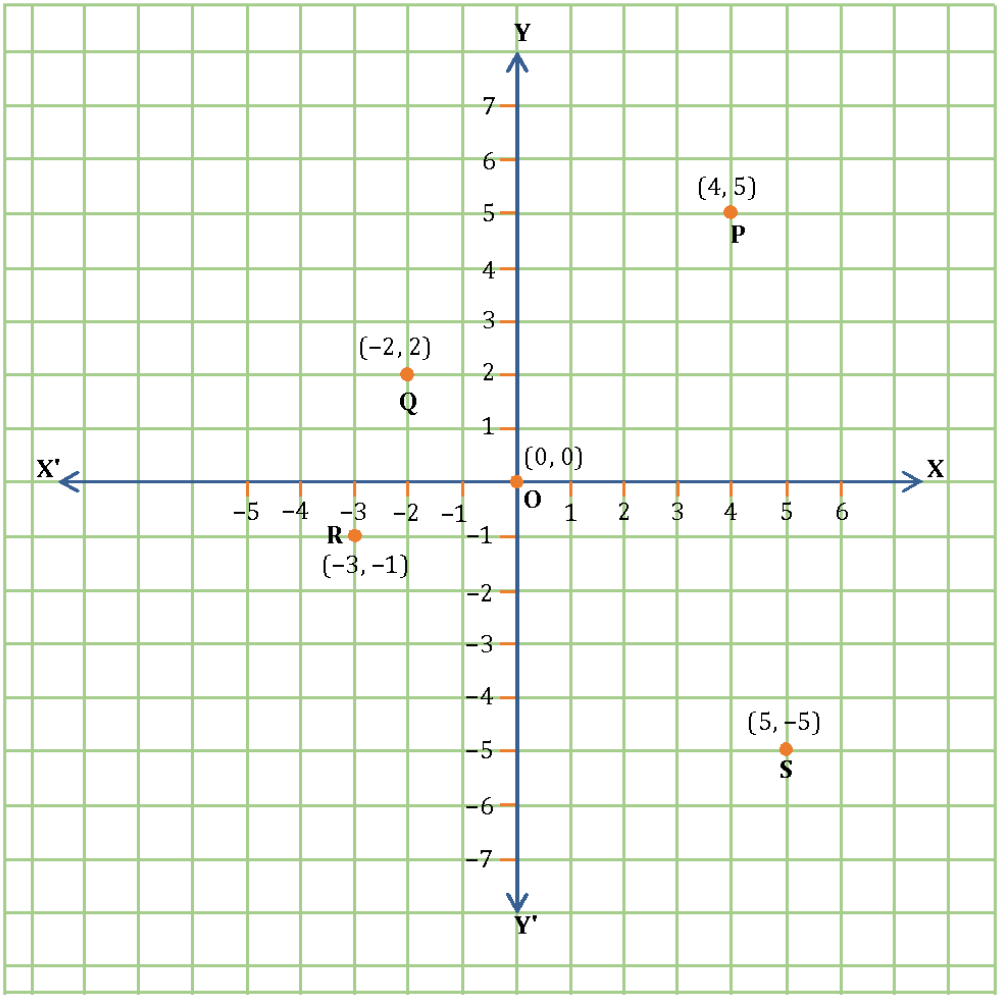
$$AB = \sqrt{(\text{Difference of abscissas})^2 + (\text{Difference of ordinates})^2}$$

SPOT LIGHT

The distance between any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

SPOT LIGHT



In the above figure, the coordinates of P be (4, 5), the coordinates of Q be (–2, 2), the coordinates be R be (–3, –1) and the coordinates of S be (5, –5).

The coordinates of origin are (0, 0)

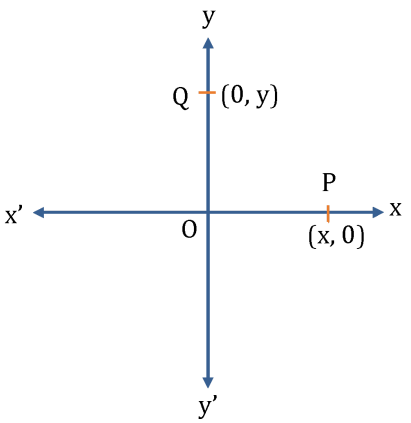
(i) For any point on x-axis, y = 0

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(ii) For any point on y-axis, x = 0

(iii) The coordinate of origin, O are (0, 0)

Points on axes



If a point P lies on **x-axis**, its coordinate is in the form of (x, 0) and if a point Q lies on **y-axis** its coordinate is in the form of (0, y) i.e., if a point P lies on x-axis its distance from x-axis will be zero. Therefore, we can say that its y-coordinate will be zero. Similarly, if a point Q lies on y-axis its distance from y-axis will be zero. Therefore, we can say that its x-coordinate will be zero.

Plotting of points

In order to plot the points in a plane, we may use the following algorithm.

Step-1 : Draw two mutually perpendicular lines on the graph paper. One horizontal and other vertical.

Step-2 : Mark their intersection point as O (origin)

Step-3 : Choose a suitable scale on x-axis and y-axis mark the points on both the axes.

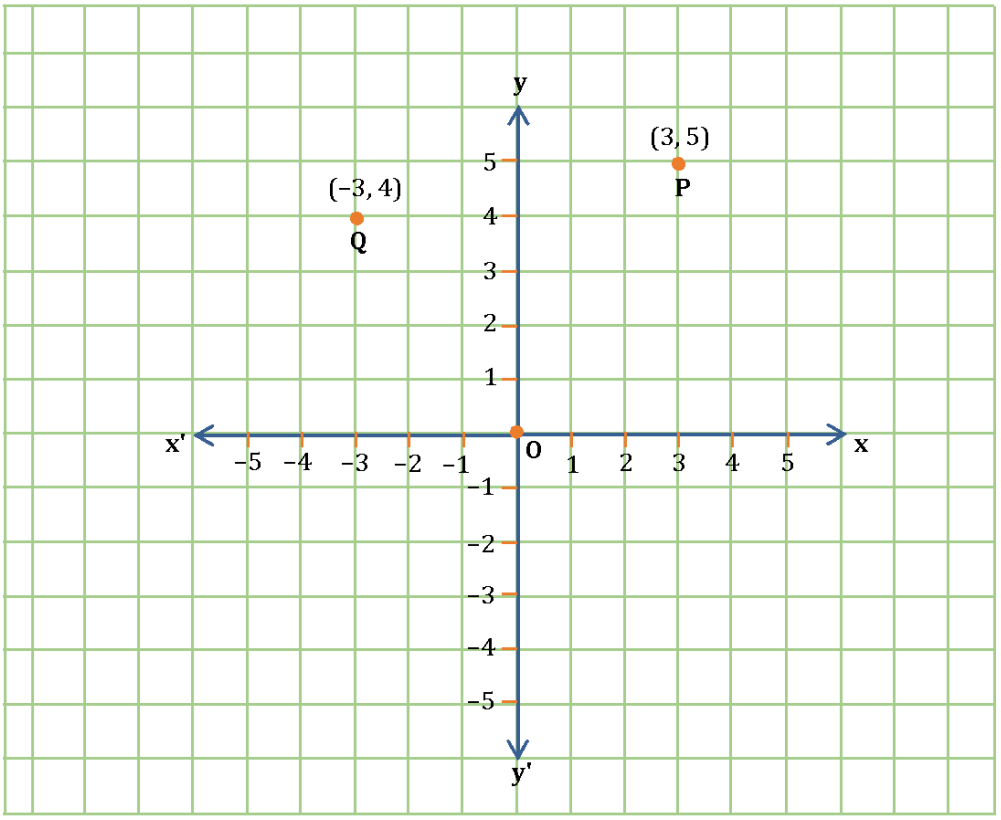
Step-4 : Obtain the coordinates of the point which is to be plotted. Let the point be P(a, b).

To plot this point start from the origin and move |a| units along ox or ox' according as a is positive or negative respectively. Suppose we arrive at point M. From point M move vertically upward or downward |b| units according as b is positive or negative. The point where we arrive finally is the required point P(a, b).

To plot a point corresponding to an ordered pair

Consider an ordered pair, say (3, 5). To plot (3, 5), we start from O, move 3 units along ox to the right of o and then 5 units upwards parallel to oy.

Similarly, to plot (–3, 4), we start from o, move 3 units along ox' and then 4 units upwards parallel to oy.



Therefore, P and Q are the required points whose coordinates are given as (3, 5) and (–3, 4) respectively.

The ordinate of a point on the x-axis is always zero. Also, abscissa of a point on the y-axis is always zero.

i. (a, 0) lies on the x-axis and (0, b) lies on the y-axis.

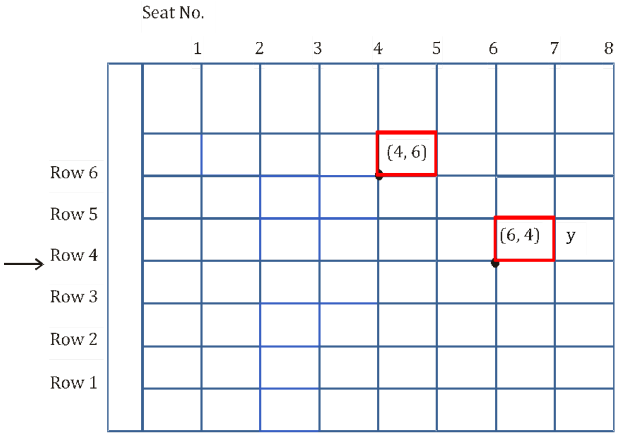
Active Maths 1

Sugam purchases a ticket for the movie "An Evening in Paris". The position of his seat is written on the ticket by using the two items of information. How will we find the position of seats in the hall.

Exploring the Concept

1. The row number is 4 in which he has to sit.
2. The seat number is 6 in that row.

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Drawing Conclusions

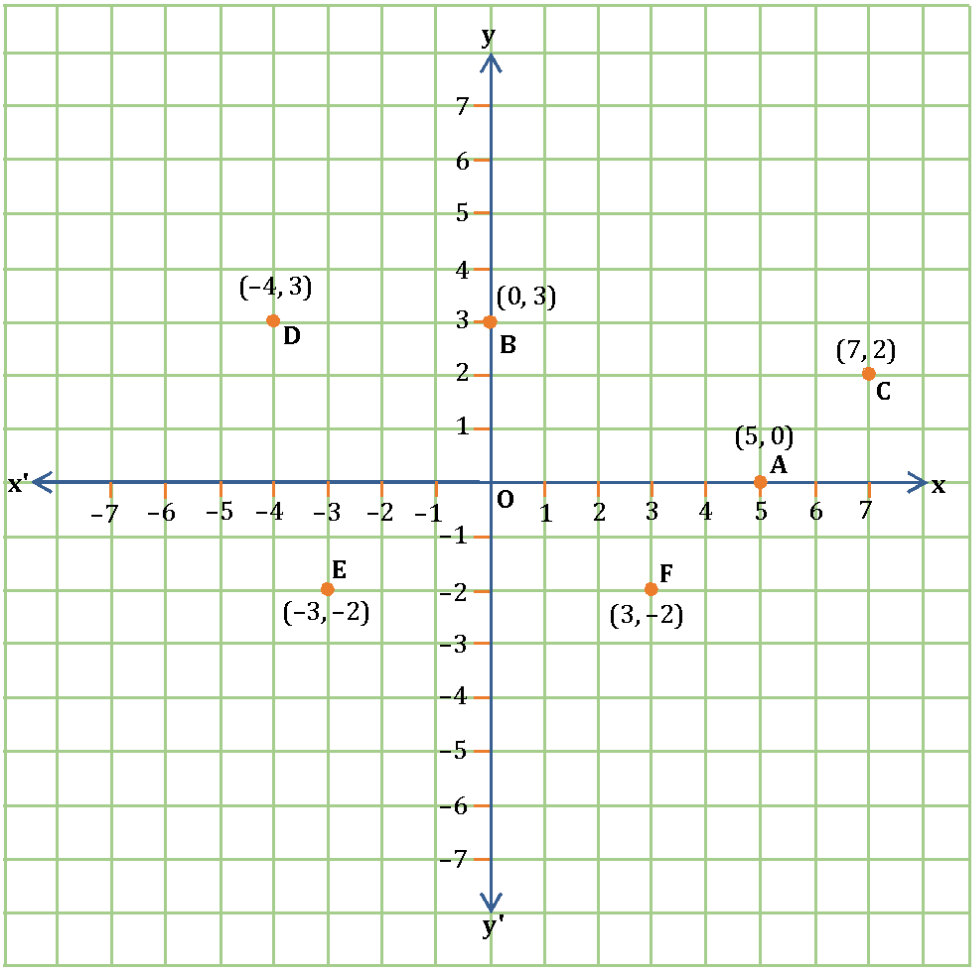
We can denote the position of this seat by (4, 6). Similarly, seat in the 6th row and seat number 4 can be written as (6, 4). Seats (4, 6) and (6, 4) are different. In this way, we can find position of any seat in the hall.

Numerical Ability 1

Locate the points in the plane if its coordinates are given as A(5, 0), B(0, 3), C(7, 2), D(–4, 3), E(– 3, –2) and F(3, – 2).

Solution

For the point A one has to move 5 units in the positive direction of the x-axis and 0 unit in the positive direction of y-axis and follow the same steps for the other points. The points on the cartesian plane are as shown in the figure.



Be Alert !

- The gap between markings on coordinate axis should be equal. Sometimes we commit this mistake by taking different lengths in between the markings.
- The orientation of co-ordinate axis is fixed (negative in left & downword and positive in right and upward).
- The numbering on axis is supposed to start of origin (intersection point of axis).

Numerical Ability 2

Plot the following pairs of numbers as points in the cartesian plane.

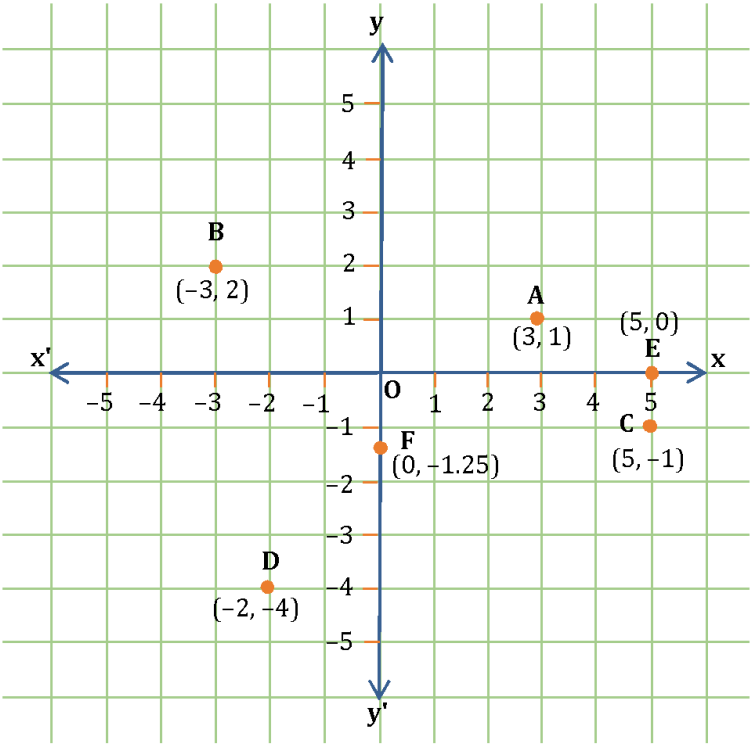
x	3	−3	5	−2	5	0

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y	1	2	–1	–4	0	–1.25
(x, y)	(3, 1)	(–3, 2)	(5, –1)	(–2, –4)	(5, 0)	(0, –1.25)

Solution

Let the points (3, 1), (–3, 2), (5, –1), (–2, –4), (5, 0) and (0, –1.25) be represented by A, B, C, D, E and F respectively.



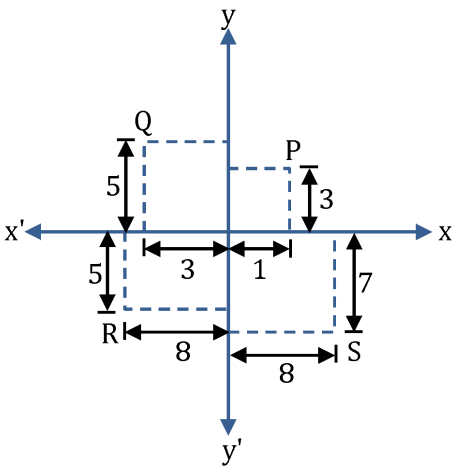
Check your Concepts 1

1. Write down the following according to given figure.

(i) Write down the abscissa of the points P, Q, R, S.

(ii) Write down the ordinate of the points P, Q, R, S.

(iii) Write down the coordinates of the points P, Q, R, S.



To find the mirror image of any figure in cartesian plane is known as reflection symmetry of that figure.

SPOT LIGHT

2. Draw a rectangle ABCD such that its vertices A, B, C and D are (4, 3), (4, –2), (–7, –2) and (–7, 3) respectively.

3. Draw a rhombus ABCD whose vertices are (1, 4.5), (–1, 0), (1, –4.5) and (3, 0) respectively.

4. Draw a triangle ABC whose vertices A, B and C are (–3, 0), (3, 3) and (–3, 3) respectively.

5. Draw a square PQRS whose vertices P, Q, R and S are (0, 0), (–4, 0), (–4, –4) and (0, –4) respectively.

6. Construct a trapezium ABCD whose vertices A, B, C and D are (3, 0), (7, 9), (–6, 9) and (–2, 0) respectively.

7. Draw a parallelogram ABCD whose vertices A, B, C and D are (–4, 8), (–4, 2), (6, –7) and (6, –1) respectively.

8. Write coordinates of a point P if its distance from x-axis is 6 units and that of from y-axis is 1 unit.

Quick Tips

It is advisable to put a small circle around the point and write the coordinates in the form (x, y)

This makes the point more clearly visible on graph paper.

Active Maths 2

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To obtain the mirror image of a given geometrical figure with respect to the x-axis and the y-axis, concept of plotting of points on the graph paper and an idea of mirror image points about a given line are needed.

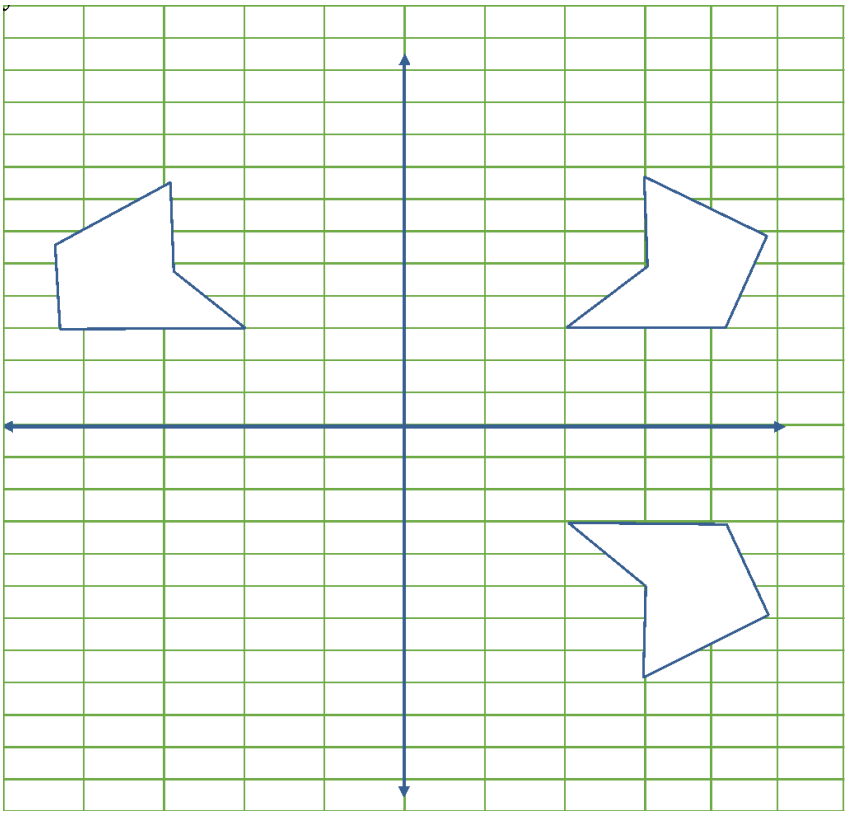
Exploring the concept

1. Consider any geometrical figure say ABCDE.

2. Plot the mirror images of the points A, B, C, D, E with respect to the x-axis to get mirror image A_1, B_1, C_1, D_1, E_1.

3. Again, plot the mirror images of points A, B, C, D, E with respect to y-axis to get mirror image A_2, B_2, C_2, D_2, E_2.

4. Repeat the activity for different geometrical figures.



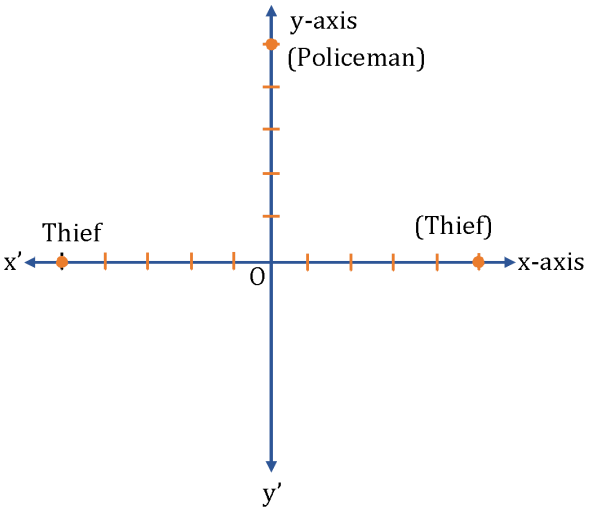
Drawing conclusions

1. The mirror images obtained with respect to the x-axis and y-axis remains same.

2. When the mirror image of a figure is obtained with respect to y-axis, then y-coordinate remains the same.

3. When the mirror image of a figure is obtained with respect to x-axis, the x-coordinate remains the same.

A policeman and a thief are equidistant from the jewellery box. Upon considering jewellery box as origin, the position of policeman is (0, 5). If the ordinate of the position of thief is zero, then the position of thief is



Explanation

Since, the ordinate of the position of thief is zero.

∴ The position of thief lies on x-axis.

Also, thief and policeman are equidistant from origin. So, position of thief is any one of (5, 0) or (– 5, 0).

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Check your Answers 1

1. (i) 1, –3, –8, 8

(ii) 3, 5, –5, –7

(iii) P(1, 3), Q(–3, 5), R(–8, –5), S(8, –7)

8. Coordinates of the point P(1, 6)

Numerical Ability 3

In which quadrant or on which axis do each of the points (–2, 4), (3, –1), (–1, 0), (1, 2) and (–3, –5) lie? Verify your answer by locating them on the cartesian plane.

Solution

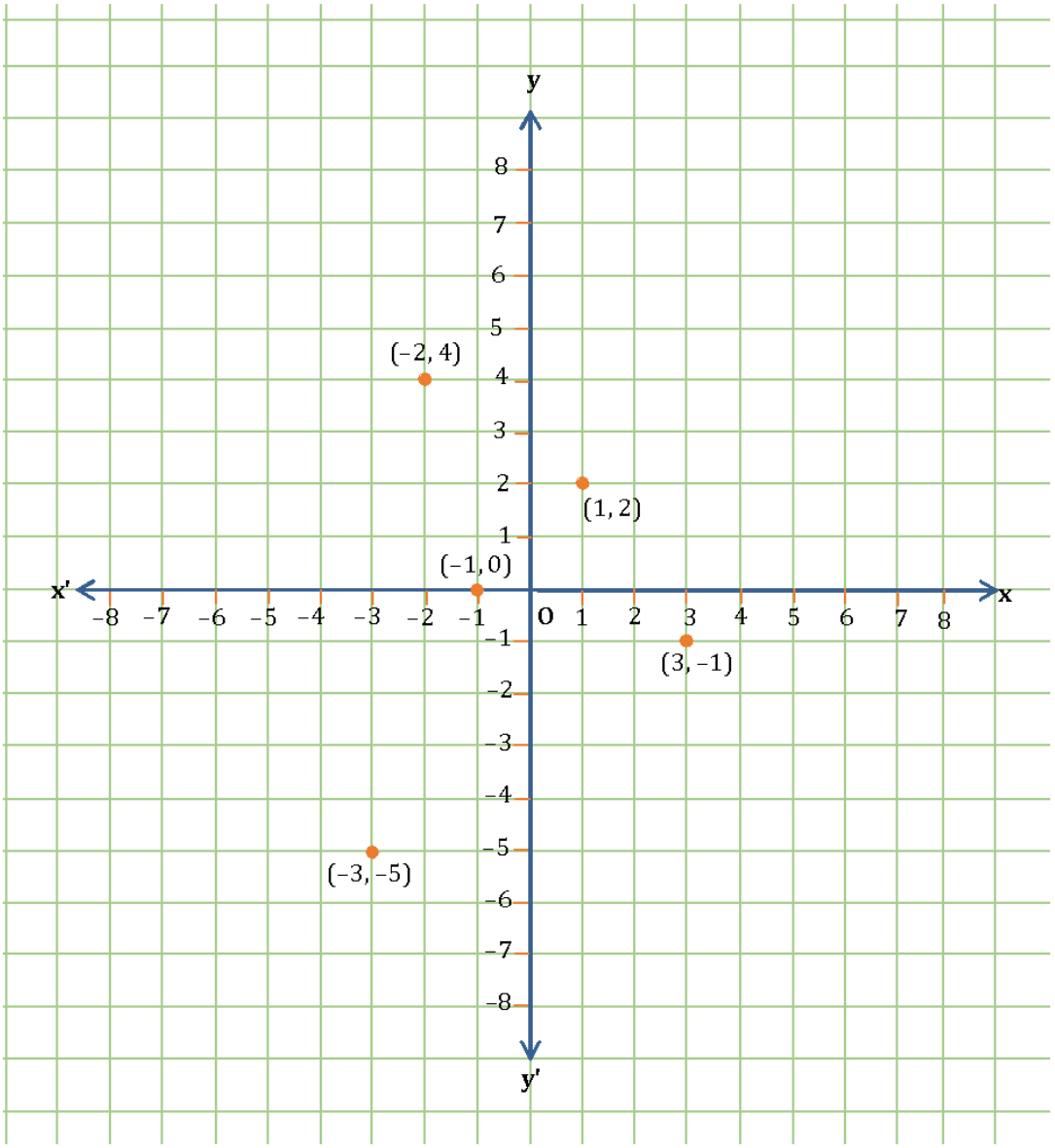
Clearly, (–2, 4) lie in the second quadrant. (x < 0, y > 0)

(3, –1) lie in the fourth quadrant. (x > 0, y < 0)

(–1, 0) lie in the x-axis and it is on the negative side of x-axis i.e. on OX'.

(1, 2) lie in the first quadrant. (x > 0, y > 0)

(–3, –5) lie in the third quadrant. (x < 0, y < 0)



Verification: Plot the points (–2, 4), (3, –1), (–1, 0), (1, 2), (–3, –5).
Result is verified.
(–2, 4) lies in II quadrant, (3, –1) lies in IV quadrant. (–1, 0) lies on left i.e. negative part of x-axis, (1, 2) lies in the I quadrant and (–3, –5) lies in the III quadrant.

Numerical Ability 4

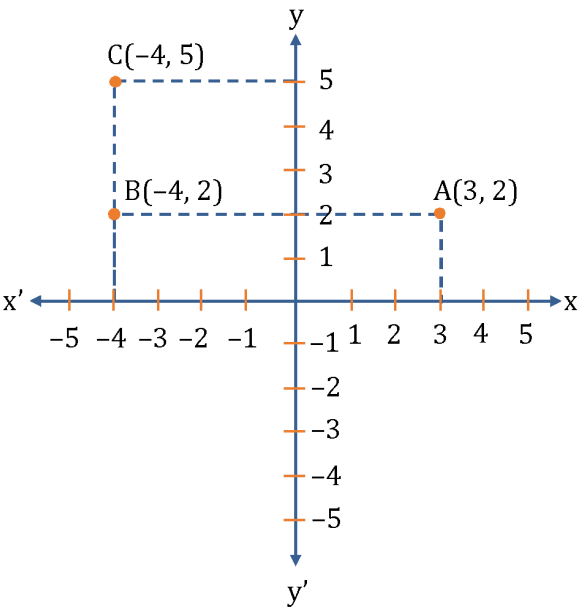
Three vertices of a rectangle are (3, 2), (–4, 2) and (–4, 5). Plot these points and find the coordinates of the fourth vertex.

Solution

Plot the three vertices of the rectangle as A(3, 2), B(–4, 2), C(–4, 5).

To find the coordinates of the fourth vertex D so that ABCD is a rectangle, the opposite sides of a rectangle are equal, therefore the abscissa of D should be equal to abscissa of A, i.e., 3 and the ordinate of D should be equal to the ordinate of C, i.e., 5.

Hence, the coordinates of D are (3, 5).



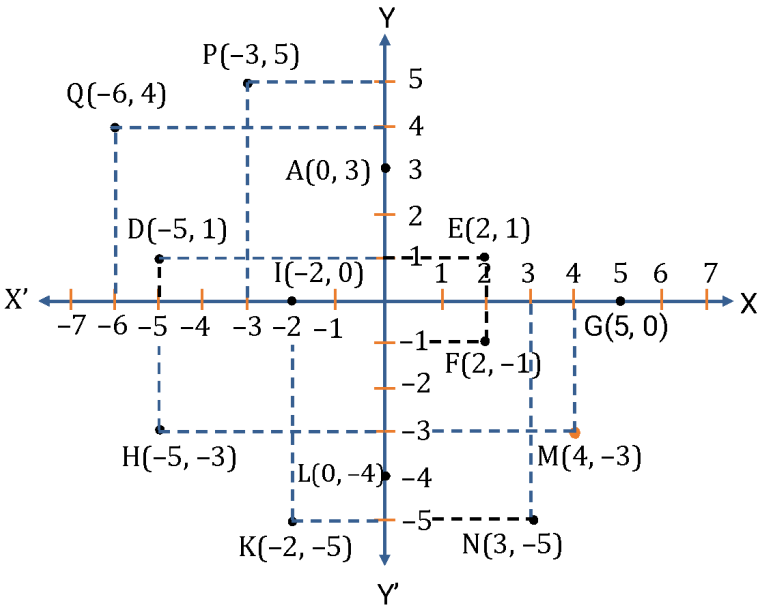
Numerical Ability 5

(a) From the figure, answer the following:

(i) Write the points whose abscissa is 0.

(ii) Write the points whose ordinate is 0.

(iii) Write the points whose abscissa is – 5.



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(b) Plot the points A(1, –1) and B(4, 5)

(i) Draw a line segment joining these points. Write the coordinates of a point on this line segment between the points A and B.

(ii) Extend this line segment and write the coordinates of a point on this line which lies outside the line segment AB.

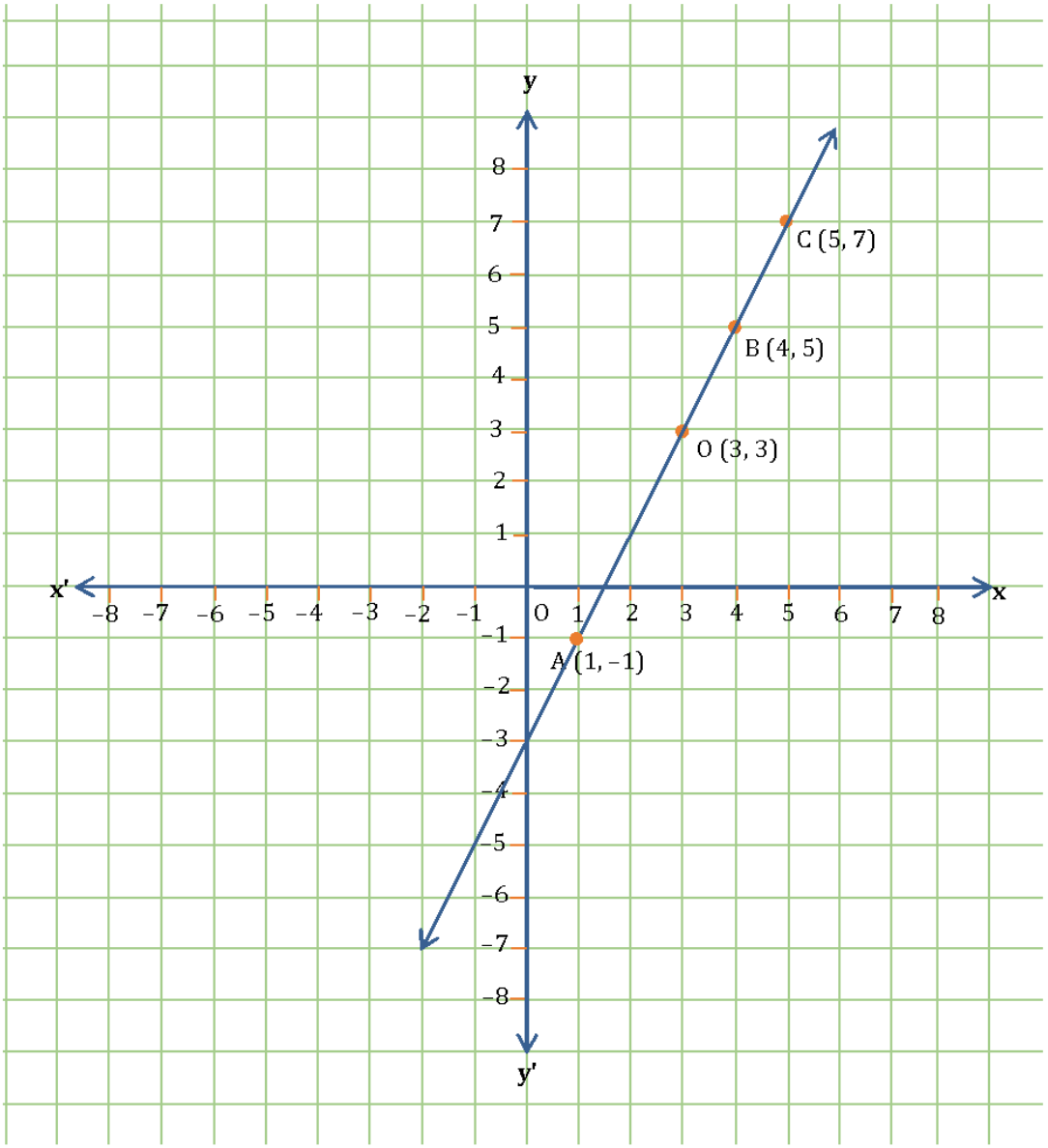
Solution

(a) (i) Required points are A(0, 3), L(0, –4)

(ii) Required points are G(5, 0), I(–2, 0)

(iii) Required points are D(–5, 1), H(–5, –3)

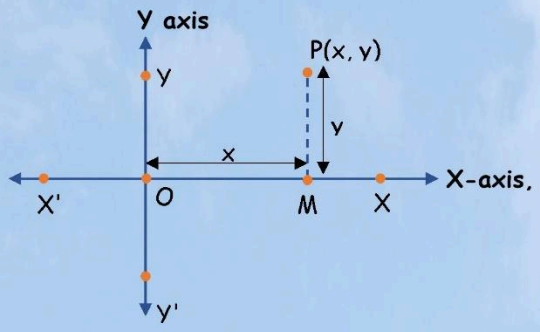
(b) Plot the points A(1, –1), B(4, 5). We can take O(3, 3) between A and B and point C(5, 7) lying outside AB but lying on AB produced. Apart from these two points, there can be many other points also.



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Memory map

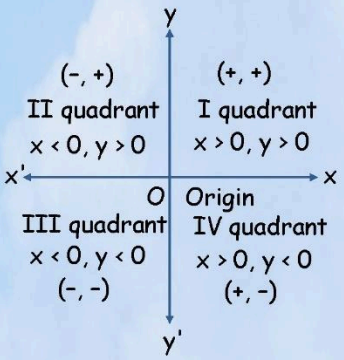
Cartesian Coordinate Axis



The above system is called a cartesian, or rectangular coordinate system.
Note that $(x, y) \neq (y, x)$

Quadrants

Region	Quadrant	Nature of x and y	Sign of the coordinates
xoy	I	$x > 0$ and $y > 0$	(+, +)
yox'	II	$x < 0$ and $y > 0$	(−, +)
x'oy'	III	$x < 0$ and $y < 0$	(−, −)
y'ox	IV	$x > 0$ and $y < 0$	(+, −)



Plotting of points

Step-1 Draw two mutually perpendicular lines on the graph paper. One horizontal and other vertical.

Step-2 Mark their intersection point as O (origin)

Step-3 Choose a suitable scale on x-axis and mark the points on both the axes.

Step-4 Obtain the coordinates of the point which is to be plotted. Let the point be P(a, b).